A Lexicalist Analysis towards Resultative Verb Constructions in Benue-Kwa Languages

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Overview

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- Lexicon rule
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- Lexicon entry for complex predicate V_1
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https://dkaramasov.github.io/mache/assets/pdf/26-a-lexicalist-analysis-towards-resultative-verb-constructions-CLUP-HO.pdf

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Resultative predicates in Germanic languages

Simpson (1983, 153 Fn.2), Boas (2003, pp. 8–11), S. M. Wechsler (2001, p. 4), S. Wechsler (2005, pp. 256–257), Müller (2002, pp. 210–211), Goldberg and Jackendoff (2004, p. 536), Beavers (2012, p. 909): Germanic resultative predicates are complex predicates, where the means predicate is a V and the result predicate is syntactically realised as AdjP or PP – but never as VP:

(1) Gaylord pulled the door [$_{AP}$ open].

(2) Gaylord pulled the door [$_{PP}$ from the truck].

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Resultative Serial Verb Constructions in Benue-Kwa

Lord (1974, 1975, pp. 24–28), Baker (1989, pp. 529–535), Déchaine (1993, p. 807), Collins (1997, pp. 472–477), Stewart (2001, pp. 149–151): Resultative Serial Verb Constructions (RSVC) consist of two V/(verbal) roots/VPs:

(3) Ó$_i$ pọ́n ___$_i$ rà.[1]
it ripen rot

'It ripened to the point of rotting.' YORÙBÁ

(4) Àsíbá$_i$ dɔ̀n [ví ɔ́]$_j$ [$_{VP}$___$_j$ yì kɔ́xɔ̀].[2]
Àsíbá pull child DET go out.side

'Àsíbá pulled the child outside.' FɔN

- Assumption here: RSVC as resultative predicates, where the result predicate is a V
- Alternative view: result predicate in Ẹ̀dó e.g. *dé* 'fall' as AdjP or PP (cf. Stewart 2001, pp. 28–29, 37)

[1] As quoted in Oyèlárăn (1982, p. 120) = ex. (19b).
[2] As quoted in Lefebvre and Brousseau (2002, p. 423) =ex. (66b).

Consequences and challenges I

- Two verbs can belong to different aspectual classes and overall RSVC
  - *fà ṣí* ‘pull open’ (YORÙBÁ): telic and gradable ACCOMPLISHMENT
  - *fà* ‘pull’: atelic ACTIVITY
  - *ṣí* ‘open’: telic and gradable ACCOMPLISHMENT/atelic STATE,
- Question how does the event composition function in detail?
- Are there any rules to determine the aspectual class of the overall RSVC?

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Consequences and challenges II

- Syntactic properties are determined by V_1 (V_1 is head)
- Aspectual properties are determined/?influenced by VP_2

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The data

- Existing data from literature
- New data from Yorùbá and Fɔn
  - former foreign language students born between 1990 and 2000
  - collected between January 2021 till March 2023 on site, via email and chats
  - elicited through translation tasks
  - cross checked through rating tasks

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Inchoative verbs, stative verbs and adjectival concepts in Benue-Kwa I

Many Benue-Kwa languages express properties and ‘adjectival concepts’ primarily as stative verbs in Yorùbá (cf. Awóbùlúyì 1967, pp. 180–182, Madugu 1976), Ìgbo (cf. Nwachukwu 1984, pp. 83–84, Nwachukwu 1987, pp. 19–20), Fɔn (cf. Lefebvre and Brousseau 2002, pp. 348–349) or Efutu (Guang) (cf. Agyeman 2016, pp. 135–142):

1. They undergo the same kind of nominalisation via partial C*í*CV-reduplication (cf. Madugu 1976, pp. 94–97)

(5) [$_{NP}$Gíga ọmọ náà] wù mí[3]
NMLZ-be.tall child the please me
‘I like the child’s tallness.’ YORÙBÁ

[3] As quoted in Madugu (1976, p. 90) = ex. (16).

Inchoative verbs, stative verbs and adjectival concepts in Benue-Kwa II

2. They take the same kind of preverbal TAM-modifiers (Madugu 1976, pp. 88–91 for Yorùbá , HABITUAL *na* in Ewe can precede a verb cf. Ameka 2008, pp. 137–139)

(6) Ọmọ náà ń ga.[4]
child DET PROG tall
'The child is getting tall.' YORÙBÁ

(7) Ọmọ náà ti ga.[5]
child DET PRF tall
'The child has become tall.' YORÙBÁ

[4] As quoted in Madugu (1976, p. 90) = ex. (12).
[5] As quoted in Madugu (1976, p. 90) = ex. (12).

Inchoative verbs, stative verbs and adjectival concepts in Benue-Kwa III

3. They participate in comparative serial verb constructions

   (8) Ọmọ náà ga ju Olu (lọ)[6]
   child the be.tall than Olu SURPASS
   'The child is taller than Olu.'

4. Yorùbá: require HIGH TONE SYLLABLE just as verb do, as well in Arigidi (Defoid) (cf. Oshòdì 2011, pp. 264, 267) and Àbèsàbèsì (Akpe, cf. Lau 2021, p. 72)
5. Ìgbo : Some of these properties require the *r*V-suffix (cf. Nwachukwu 1984, pp. 83–84)

[6] As quoted in Madugu (1976, p. 90) = ex. (17).

Inchoative verbs, stative verbs and adjectival concepts in Benue-Kwa IV

Most of these languages small(er) set of ‘closed class’ more marked items: e.g. 6 for Ìgbo (cf. Nwachukwu 1984, pp. 97–99), 18 for Fɔn (cf.Lefebvre and Brousseau 2002, pp. 350–352):

1. They occur adnominal as attributes (cf. Nwachukwu 1984, pp. 97–99 for Ìgbo , Lefebvre and Brousseau 2002, pp. 350–352 for Fɔn , Agyeman 2016, pp. 137–138 for Efutu)

(9) [$_{NP}$nwókó ọ́cha.]
man PFX-be.light.colored?

‘a fair complexioned man’ ÌGBO

‘Genuine’ adjectival concepts in Benue-Kwa II

2. Do not participate in C*í*CV-reduplication, as they are reduplicated forms or nominalisations already (cf. Madugu 1976, pp. 94–97)

(10) [$_{NP}$ọmọ kékéré/dúdú.]
child RED-small/black

‘a small/black child’ YORÙBÁ

3. Lefebvre and Brousseau (2002, p. 351) ‘genuine’ adjectives cannot occur as the main predicate in Fɔn , but require a copula, similar in Efutu (cf. Agyeman 2016, pp. 137, 171–172)

(11) a. * Kɔ̀kú ɖàgbè / ɖàgbó / vú / glì.
Kɔ̀kú good / important / young / short
Intended: ‘Kɔ̀kú is good/important/young/short.’
FɔN

b. Kɔ̀kú ɖò ɖàgbè / ɖàgbó / vú / glì.
Kɔ̀kú be.at good / important / young / short

Verbs and gradability

In some languages verbs can express gradable states and events: they take degree intensifiers or occur in *wie*-exclamatives

(12) Sie hat sich sehr/extrem gefreut.
she has REFL much/extremely be.happy
'She was very/extremely happy.'

(13) Wie (sehr) der Helmerl heute wieder auszuckt.
how much DET Helmerl today again get.mad
'How much Helmerl got upset again today!'

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Tense and Aspect in Benue-Kwa

	BK1 (old) Àkán , Ẹ̀dó , Ìgbo , Bantu, . . .	BK2 (new) Gbè, Yorùbá , . . .
PAST/PRESENT ambiguity	—	✓
only one event per clause	—	✓
pro-clitic inflection	—	✓
three surface tones	—	✓

Figure: Two large groups of languages among Benue-Kwa according to Manfredi (2005, p. 2)

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The interpretation of unmarked/tenseless verbs in BK 2: The FACTATIVE

Benue-Kwa 2 languages make use of unmarked finite verb forms such, as Yorùbá (cf. Awóbùlúyì 1967, pp. 260–261, 265, Welmers 1973, pp. 346–347 and Awóyalé 1991, p. 218) and Gungbe (cf. Aboh and Nauze 2008, pp. 217–218) with factative interpretation:

- Default interpretation of eventive predicates: past tense
- Default interpretation of stative predicates: present tense
- Some telic ACHIEVEMENT/ACCOMPLISHMENT verbs are ambiguous between a PRESENT and a PRESENT PERFECT interpretation (cf. Avolonto 1992, pp. 97–98 and Lefebvre and Brousseau 2002, pp. 86–89 for Fɔn)

Extends to Western Kwa: Efutu (Guang), as illustrated by Agyeman (2016, pp. 186–190)

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The interpretation of unmarked/tenseless verbs in BK 2: Alternative view

Essegbey (1999, pp. 32, 64–67) and Ameka (2008, pp. 135–136, 139–140) argue for Ewe-Gbè: Only two types of verb classes:

1. “Action verbs” (≈ ACTIVITIES)
2. Inchoatives (≈ ACHIEVMENTS and ACCOMPLISHMENTS): Stative verbs can be systematically derived from inchoatives

Plausible analysis for Yorùbá inchoatives and statives (cf. Manfredi pers. commun)

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Inchoative and stative interpretations

(14) a. Òkè náà ga.[7]
mountain DET tall

‘The mountain is high.’ YORÙBÁ

b. Òkè náà ń ga.
mountain DET PROG tall

‘The mountain is becoming high.’ YORÙBÁ

#‘The mountain is being high.’

- Two possible conclusions:
  1. ‘Stative’ predicates can be coerced into inchoative interpretations
  2. Or: inchoative is the base form which only becomes visible in certain contexts
- Assumption here: Option 2; unmarked verbs have default interpretations, but stative interpretations always derived from eventive ones

[7] Elicited examples (translation task).

The interpretation of tensed verbs in BK 1: Ìgbo

All eventive and most stative predicates in Ìgbo are inflected with the so-called (FACTATIVE) *r*V-suffix (cf. Welmers 1973, pp. 346–347 and Manfredi 1991, p. 174).

(15) a. * Ógu tóro.[8]
Ógu tall-rV
Intended: ‘Ógu is/became tall. ÌGBO

b. Ógu tóro eto
Ógu tall-rV NMLZ-tall
‘Ógu is/became tall. ÌGBO

- Default interpretation with most stative verbs: present tense.
- Default interpretation with eventive verbs: past tense.
- Inchoatives mostly derive a stative interpretation
- The *r*V-suffix can never be clause final: intransitive verbs have BOUND VERB COMPLEMENT

[8] As quoted in Nwachukwu (1987, pp. 19–20) = ex. (12a–b).

The interpretation of tensed verbs in BK 1: Ẹ̀dó

Manfredi (2022, p. 5): Also Ẹ̀dó predicates can remain ambiguos between a inchoative/telic interpretation and a stative interpretation in the presence of the (FACTATIVE) (*r*)-ẹ̀ suffix:

(16) a. Ọ̀ gọ́-(r)ẹ̀.[9]
3S.S bend-FACT

‘It became bent.’ ẸDÓ

‘It is [now] bent.’

b. Ọ̀ guọghọ́-(r)ẹ̀.[10]
3S.S break-FACT

‘It broke.’ ẸDÓ

‘It is [now] broken.’

[9] As quoted in Manfredi (2022, p. 5) = ex. (25a).
[10] As quoted in Manfredi (2022, p. 5) = ex. (25b).

Resultative Serial Verb Constructions – Yorùbá

jẹ ⟨AG, TH⟩ ‘eat’ + *tán* ⟨TH⟩ ‘be.finish’ → *jẹ* X *tán* ⟨AG, TH⟩ ‘eat X be.finish’

(17) Ẹniọlá á jẹ [ọbẹ̀ ewédú] tán.
Ẹniọlá HTS eat soup jute.leaf be.finished
‘Ẹniọlá finished the ewédú.’ (SVC)

a. Ẹniọlá á jẹ [ọbẹ̀ ewédú].
Ẹniọlá HTS eat soup jute.leaf
‘Ẹniọlá ate ewédú.’ (TRANS)

b. Ọbẹ̀ ewédú ú tán.
soup jute.leaf HTS be.finished
‘The ewédú soup is finished.’ (UNACC)

- HIGH TONE SYLLABLE precedes V_1 and mostly limited to veridical contexts

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Syntactic headedness

- Imperative possible with full RSVC

(18) Jẹ ọbẹ ewédú tán!
eat soup jute.leaf be.finished
'Finish up the ewédú!' YORÙBÁ

- Imperative possible with V_1

(19) Jẹ ọbẹ ewédú!
eat soup jute.leaf
'Eat the ewédú!' YORÙBÁ

- Sometimes imperative impossible with V_2

(20) # tán!
be.finished
Intended:'Be finished!' YORÙBÁ

- V_1 is the syntactic head, similar conclusion by Ogie (2009, pp. 476–479)

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Syntactic microvariation within Benue-Kwa I

Largely absent from Gbè-Cluster, Ka-Togo and Na-Togo, as illustrated for FɔN -Gbè

(21) Kɔ̀kú$_i$ xò [kòklô ɔ́]$_j$ __$_{i/*j}$ hù.[11]
Kɔ̀kú hit chicken DEF kill
‘ Kɔ̀kú beat the chicken to death.’ FɔN

(22) * Kɔ̀kú$_i$ xò [kòklô ɔ́]$_j$ __$_j$ kú.[12]
Kɔ̀kú hit chicken DEF die

[11] As quoted in Lefebvre and Brousseau (2002, p. 414) = ex. (41).
[12] As quoted in Lefebvre and Brousseau (2002, p. 415) = ex. (42).

Syntactic microvariation within Benue-Kwa II

- Same for other Gbè-languages: εuε Ameka (2003, pp. 9–12; 2005, pp. 361–364, 373), Gun (cf. Aboh 2009, pp. 21–23)
- Similar data from Ka-Togo: Avatime (cf. Defina 2016, p. 100), Tutrugbu (cf. Essegbey 2019, pp. 245–246), Tuwuli (cf. Harley 2005, pp. 432–451)
- Na-Togo: Likpe (cf. Ameka 2003, p. 11)
- Outside Benue-Ka: Dàgáárè (Gur, Hiraiwa and Bodomo 2008, pp. 808–809)

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Aspectual properties of resultative predicates in Germanic

- S. M. Wechsler (2001, p. 4, 2005, p. 258): Canonical resultative predicates are always telic
- S. M. Wechsler (2001, pp. 9–12, 2005, pp. 264–268) and Beavers, p. 925 (2012, p. 925, 2013, pp. 690–692) two large subtypes of resultative predicates
  1. ACCOMPLISHMENTS: durative V + gradable closed-scale adjective, e.g. *Mary hammered the metal flat*
  2. ACHIEVEMENTS: punctual V + non-gradable, closed-scale maximum endpoint Adj, e.g. *He shot the miller dead*

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Modeling telicity according to Beavers (2013, pp. 684–688)

(23) a. John warmed the pie to 100° in/?for an hour.

b. $\exists e \exists s[\textit{warming}'(\mathbf{j}, \mathbf{p}, s, e) \wedge \textit{result}'(s, \mathbf{100°}, e)]$

(24) a. John warmed the pie for/??in an hour.

b. $\exists e \exists g \exists s[\textit{warming}'(\mathbf{j}, \mathbf{p}, s, e) \wedge \textit{result}'(s, g, e)]$

- Telic predicates (quantised, cf. 23) come with specified goal arguments *g* and a simple/complex scale *s*.
- Some atelic predicates (non-quantised, cf. 24) have unspecified goal arguments *g*
- There is a relation which defines *g* as the result of the event *e*: result(s,g,e)

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Aspectual classes of RSVC in Benue-Kwa

Ogie (2009, pp. 235–238): Benue-Kwa also has two types of RSVC: gradable and durative ACCOMPLISHMENTS (cf. 25) and punctual non-gradable ACHIEVEMENTS (26)

(25) Òzó$_i$ họ̀ọ́ úkpọ̀n$_j$ __$_j$ huán.[13]
Òzó wash.PST cloth be.clean.PST
'Òzó washed the clothes clean.' ỊDÓ

(26) Òzó suá Àzàrí [$_{VP}$dé gbé òtò] vbé
Òzó push.PST Àzàrí fall.PST against ground at
ágógó èvá/ *ífuánrò évá.[14]
clock two minute two
'Òzó pushed Àzàrí down at 2pm/*in two minutes.'

[13] As quoted in Ogie (2009, p. 236) =ex. (69a).
[14] As quoted in Ogie (2009, p. 237) =ex. (70b).

Atelic readings with RSVC

Challenge: Durative adverbs may coerce atelic readings with RSVC: goal will be interpreted as unspecified on the content value of the modifier.

(27) Olú ti ọmọ náà ṣubú fún wákàtí kan.[15]
Olú push child DET fall give hour one
'Olú pushed the child for one hour' (ITERATIVE)
'Olú pushed the child for one hour' (CONTINUOUS)

(28) Fẹ́mi í fọ̀ aṣọ náà mọ́n fún wákàti kan.[16]
Fẹ́mi HTS wash cloth DET clean for hour one
'Fẹ́mi washed the cloth clean/thoroughly for an hour.'
(CONTINUOUS)

[15] Elicited example (translation task).
[16] Elicited example (translation task).

Previous work: Derivational Approaches

Analysis of interpretation of the whole RSVC: Stewart (2001, p. 27), Zimmermann and Amaechi (2025)

(29) a. Òzó sùá ọ̀gọ́ dé.[17]
Òzó push bottle fall
'Òzó pushed the bottle down.' ẸDÓ

b. (∃*e*) [Push-Fall(e) & AGENT(e, Òzó) & THEME(e,ọ̀gọ́)]

(30) a. Ó gbù-dà-rà ósísí.[18]
3S.S chop-fall-rV tree
'He felled the tree.' ÌGBO

b. ∃*e*∃*s* [AG(Maria,e) ∧ chop(e) ∧ TH(tree,e) ∧ Caus(e,s) fallen(s) ∧ HOLD(tree,s)]

[17] As quoted in Stewart (2001, p. 27) = ex. (14a–b).
[18] As quoted in Zimmermann and Amaechi (2025, p. 45) = ex. (93a–b).

Previous work: Type-Logical Grammar

Naumann and Gamerschlag (2021, pp. 354–355) provide a constructor rule which built a RSVC from two simple predicates (31) and an example of the type *push* X *fall* (32).

(31) $\lambda P_1.\lambda P_2.\lambda x.\lambda y.\exists e_1.\exists e_2[P_1(e_1) \wedge P_2(e_2) \wedge \textit{arg-pattern}(e_1, e_2), x, y) \wedge \textit{relation}(e, e_1, e_2)]$[19]

(32) $\lambda x.\lambda y.\lambda e.\exists e_1.\exists e_2[\text{push}(e_1) \wedge \text{fall}(e_2) \wedge \text{actor}(e_1) = x \wedge \text{theme}(e_1) = y = \text{theme}(e_2) \wedge e = e_1\ e_2 \wedge \textit{cause}(e_1, e_2)\ ]$
[20]

A further analysis within HPSG/Minimal Recursion Semantic for Ẹ̀dó is provided by Ogie (2009, pp. 477–480), mostly occupied with argument identification.

[19] As quoted in Naumann and Gamerschlag (2021, p. 354) = ex. (35).

[20] As quoted in Naumann and Gamerschlag (2021, p. 355) = ex. (37).

RSVC in Benue-Kwa II

Ingredients for the analysis for object-oriented RSVC

1. Lexicon rules transforming V_{trans} $\langle NP_{AG}, NP_{TH} \rangle$ into $V_{trans\text{-}res}$ selecting for $\langle NP_{AG}, NP_{TH}, Pred \rangle$
   - event composition: *macroevent e* is a String Type composed of concatenation of events expressed by the verbal components (cf. Bohnemeyer, Enfield, et al. 2007, pp. 497, 502–508 and Bohnemeyer and Van Valin 2017, pp. 144–148, Fernando 2004, . . .)
   - goal argument *g* inherited from the result predicate VP_2: contextually given *g* necessary for a telic interpretation
   - scale ζ inherited the result predicate VP_2: determines the type of scale: simplex/complex
   - Following Beavers (2013, pp. 684–688), there is the a relation which defines *g* as the result of the event *e*: result(ζ,g,e_2)
2. phrasal immediate dominance schemes combining complex predicate with predicate argument

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Event Structure Templates

Van Valin and Lapolla (1997, pp. 102–113) and Rappaport Hovav and Levin (1998, pp. 109–111): UG provides small number of universal event structure templates

(33) a. ACTIVITIES: [*x* ACT$_{\langle MANNER \rangle}$ (*y*)]

b. STATES: [*x* ⟨*STATE*⟩]

c. ACHIEVEMENTS: [BECOME [*x* ⟨STATE⟩]]

d. ACCOMPLISHMENTS: [*x* CAUSE [BECOME [z ⟨STATE⟩]]]

e. ACCOMPLISHMENTS: [[*x* ACT$_{\langle MANNER \rangle}$ (*y*)] CAUSE [BECOME [*z* ⟨STATE⟩]]]

- Rappaport Hovav and Levin (1998, pp. 97, 123, 127–130) projected into syntax from the lexicon entry of the V
- based on four aspectual classes
- cover all verb meanings

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Template Augmentation rules

This can be ensured by means of a formalisation of *Template Augmentation Rule* inspired by Rappaport Hovav and Levin (1998, pp. 118–123)

- Lexical rule which takes a transitive ACTIVITY V_1 as input and returns a resultative transitive ACCOMPLISHMENT V_1
- Adds an additional intransitive ACHIEVEMENT V_1 verb on its ARG-ST list of V_1
  turns ACHIEVEMENT predicates in possible result states, which lack the MEP and therefore qualify as V_2 in a RSVC
- This V_2 has a defective event structure lacking the macro event property
- Complex ACCOMPLISHMENT V_1 only denotes a single complex event

Cooper (2005, p. 99), Cooper (2023) attempt to fuse several theories into one formalism:

1. Montague Semantics (λ-calculus)
2. Situation Semantics (following tradition of Jon Barwise)
3. Discourse Representation Theory
4. HPSG
5. Probablistic semantics
6. Incremental processing/Dynamic Syntax
7. Semantic frames

Analysis here implemented in *Conversation Oriented Semantics* (*KoS*) as proposed by Ginzburg (2012)

Situation Semantics: Situation type corresponds to the set of possible worlds consistent with the state affairs described by the situation: eg. worlds in which someone hugs someone (cf. 34a), or a boy hugs a dog (cf. 34a):

(34) a.
$$\begin{bmatrix} x & : & \text{Ind} \\ y & : & \text{Ind} \\ e & : & \text{hug(x,y)} \end{bmatrix}$$

b.
$$\begin{bmatrix} x & : & \text{Ind} \\ c_1 & : & \text{boy(x)} \\ y & : & \text{Ind} \\ c_2 & : & \text{dog(y)} \\ e & : & \text{hug(x,y)} \end{bmatrix}$$

c. $\exists x\,[\text{boy}(x) \wedge \exists y\,[\text{dog}(y) \wedge \text{hug}(x,y)]]$

TTR assumes a rich ontology of types, which are represented as Record Types.

(35) a. $\langle e, \langle e, t \rangle \rangle$

b. $\lambda x \lambda y . [hug'(x, y)]$

c. $\lambda r_1{:}[x^c{:}\mathrm{IND}].\ \lambda r_2{:}[x^s{:}\mathrm{IND}].$

$$\left[\ c_1 \ : \ \mathrm{hug}(r_2.x^s, r_1.x^c)\ \right]$$

d. $\lambda r_1{:}[x^c{:}\mathrm{IND}].\ \lambda r_2{:}[x^s{:}\mathrm{IND}].$

$$\left[\begin{array}{lcl} r_2.x^s & : & \mathrm{IND} \\ r_1.x^c & : & \mathrm{IND} \\ c_1 & : & \mathrm{hug}(r_2.x^s, r_1.x^c) \end{array}\right]$$

Ingredients

- Lexicon entries with rich event structures
- Lexicon rule that turns transitive ACTIVITY V_1 into ACCOMPLISHMENT V_2, adding extra defective resultative ACHIEVEMENT predicate V_2 to ARGUMENT STRUCTURE list
- Lexicon rule that turns common achievement predicate into V_2
- V_2 is slightly grammaticalised lacking MEP

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Event structure: ACTIVITY

$$\lambda r_1{:}[x^c{:}\mathit{Ind}].\ \lambda r_2{:}[x^s{:}\mathit{Ind}].\ \begin{bmatrix} r_2.x^s{:}\mathit{Ind} \\ r_1.x^c{:}\mathit{Ind} \\ e{=}e\text{-}str.e{:}\mathit{Event} \\ e\text{-}str: \begin{bmatrix} x{:}\mathit{Ind} \\ y{:}\mathit{Ind} \\ e_1{:}\mathit{Event} \\ e{=}e_1{:}\mathit{Event} \\ \zeta{=}\ \neg\mathrm{Boolean}{:}\mathit{Scale} \\ c_{MSTATE}{:}\mathrm{drink}(x,y,\zeta,e_1) \\ c_{ACT}{:}\mathrm{do}(x,c_{STATE},e_1) \end{bmatrix} \end{bmatrix}$$

- More specified notation of $\lambda y.\lambda x.\mathrm{drink}(x, y)$
- First line specifies the types of argument: $2\times$ *Ind* (e)
- Second line specifies the $+$MEP
- Third line describes event structure

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Event structure: ACCOMPLISHMENT

$$\lambda r_1{:}[x^c{:}\mathit{Ind}].\ \lambda r_2{:}[x^s{:}\mathit{Ind}].\ \begin{bmatrix} r_2.x^s{:}\mathit{Ind} \\ r_1.x^c{:}\mathit{Ind} \\ e{=}e\text{-}str.e{:}\mathit{Occur} \\ e\text{-}str{:} \begin{bmatrix} x{:}\mathit{Ind} \\ y{:}\mathit{Ind} \\ e_1{:}\mathit{Event} \\ e_2{:}\mathit{Event} \\ e{=}e_1 \frown e_2{:}\mathit{Occur} \\ \zeta{=}\neg\text{Boolean}{:}\mathit{Scale} \\ c_{\text{MSTATE}}{:}\text{drink}(x,y,\zeta,e_1) \\ c_{\text{ACT}}{:}\text{do}(x,c_{\text{STATE}},e_1) \\ c_{\text{BECOME}}{:}\text{become}(c_{\text{RSTATE}},\zeta) \\ c_{\text{RSTATE}}{:}\neg\text{exist}(x,y,\zeta,e_2) \end{bmatrix} \end{bmatrix}$$

- More specified notation of $\lambda y.\lambda x.\text{drink.up}(x, y)$
- First line specifies the types of argument: $2\times$ *Ind* (e)
- Second line specifies the $+$MEP
- Third line describes event structure

CLUL

Lexical rule for *Template Augmentation*: Function

- Takes an transitive ACTIVITY V as input
- Returns a transitive ACCOMPLISHMENT V as output
- Adds a defective resultative predicate to argument list
- Resultative predicate is an intransitive ACHIEVEMENT
- Resultative predicate is defective lacks MEP
- Output event structure is merger of empty ACCOMPLISHMENT template with the ACTIVITY V_1 and the ACHIEVEMENT V_2

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Lexical rule for *Template Augmentation*

$$\textbf{vrsvc}: \left[\begin{array}{l} \text{arg-st}=\boxed{1}\oplus\boxed{2}\oplus\langle\, v_2{:}\left[\begin{array}{l}\text{cat}:\left[\begin{array}{l}\text{head}=\text{V}:\mathit{PoS}\\ \text{subj}=\boxed{2}:\mathit{PoS}\end{array}\right]\\ \text{arg-st}=\langle\boxed{2}\rangle:\mathit{list(synsem)}\end{array}\right]\rangle:\text{list(synsem)} \\ \text{cont}=\lambda r_1{:}\left[\begin{array}{l}\text{e}=\text{e-str.e}_2:\mathit{Event}\\ \text{e-str}=\boxed{4}:\left[\begin{array}{l}\text{y}=\boxed{2}:\mathit{Ind}\\ \text{e}_2:\mathit{Event}\\ \zeta=\neg\text{Boolean}:\mathit{Scale}\\ \text{c}_{\text{BECOME}}:\text{become}(\text{c}_{\text{RSTATE}},\zeta)\\ \text{c}_{\text{RSTATE}}:\text{stpredicate}(\text{y},\zeta,\text{e}_2)\end{array}\right]\end{array}\right].\ \lambda r_2{:}[\text{y}^c=\boxed{2}:\mathit{Ind}].\ \lambda r_3{:}[\text{x}^s=\boxed{1}:\mathit{Ind}]. \\ \quad\left[\begin{array}{l}\text{e}=\text{e-str.e}:\mathit{Occur}\\ \text{e-str}:\text{Accomplishment}\dot{\wedge}\boxed{3}\ \dot{\wedge}\boxed{4}\end{array}\right]:\mathit{DTV} \end{array}\right]$$

input-dtr

$$\textbf{tv-dtr}: \left[\begin{array}{l}\text{arg-st}=\boxed{1}\langle\text{np}_1{:}\left[\text{x}^s=\text{cont.x}^s:\mathit{Ind}\right]\rangle\oplus\boxed{2}\langle\text{np}_2{:}\left[\text{y}^c=\text{cont.y}^c:\mathit{Ind}\right]\rangle:\mathit{list(synsem)}\\ \text{cont}=\lambda r_2{:}[\text{y}^c=\boxed{2}:\mathit{Ind}].\lambda r_3{:}[\text{x}^s=\boxed{1}:\mathit{Ind}].\\ \quad\left[\begin{array}{l}\text{x}^s=\boxed{1}:\mathit{Ind}\\ \text{y}^c=\boxed{2}:\mathit{Ind}\\ \text{e}=\text{e-str.e}_1:\mathit{Event}\\ \text{e-str}=\boxed{3}:\mathit{Activity}\end{array}\right]:\mathit{TV}\end{array}\right]$$

Relabelling rule for defective ACHIEVEMENT predicate: Function

- Takes an intransitive ACHIEVEMENT V with MEP
- Returns a defective ACHIEVEMENT without MEP
- Relabels subject argument *x* as object argument *y*

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Relabelling rule for defective ACHIEVEMENT predicate

$$\textbf{intr-obj-rel}: \left[\begin{array}{l} \text{cont}= \lambda r_1{:}[\text{y}^s{:}\mathit{Ind}]. \left[\begin{array}{l} r_1.\text{y}^s{=}\text{e-str.y}{:}\mathit{Ind} \\ \text{e}{:}\mathit{Event} \\ \text{e-str}: \left[\begin{array}{l} \text{y}{=}\text{intr-dtr.cont.e-str.x}{:}\mathit{Ind} \\ \text{e}_2{=}\text{intr-dtr.cont.e-str.e}_1{:}\mathit{Event} \\ \zeta{=}\text{intr-dtr.cont.e-str.}\zeta{=}\neg\text{Boolean}{:}\mathit{Scale} \\ \text{c}_{\text{BECOME}}{:}\text{become}(\text{c}_{\text{RSTATE}},\zeta) \\ \text{c}_{\text{RSTATE}}{:}\neg\text{exist}(\text{x},\text{y},\zeta,\text{e}_2) \end{array}\right] \end{array}\right] {:}\mathit{IV} \\ \qquad\qquad \text{input-dtr} \\ \qquad\qquad \Big| \\ \textbf{intr-dtr}: \left[\begin{array}{l} \text{cont}=\lambda r_1{:}[\text{x}^s{:}\mathit{Ind}]. \left[\begin{array}{l} r_1.\text{x}^s{=}\text{e-str.x}{:}\mathit{Ind} \\ \text{e}{=}\text{e-str.e}{:}\mathit{Event} \\ \text{e-str}: \left[\begin{array}{l} \text{x}{:}\mathit{Ind} \\ \text{e}_1{:}\mathit{Event} \\ \text{e}{=}\text{e}_1{:}\mathit{Event} \\ \zeta{=}\neg\text{Boolean}{:}\mathit{Scale} \\ \text{c}_{\text{BECOME}}{:}\text{become}(\text{c}_{\text{RSTATE}},\zeta) \\ \text{c}_{\text{RSTATE}}{:}\neg\text{exist}(\text{x},\text{y},\zeta,\text{e}_1) \end{array}\right] \end{array}\right] {:}\mathit{IV} \end{array}\right]$$

Figure: Relabelling Rule for object-oriented intransitive ACHIEVEMENT predicates

Event structure: ACHIEVEMENT

$$\lambda r_1{:}[x^s{:}\mathit{Ind}].\begin{bmatrix} r_1.x^s{=}x.\text{e-str}{:}\mathit{Ind} \\ e{=}\text{e-str}.e{:}\mathit{Event} \\ \text{e-str}{:}\begin{bmatrix} x{:}\mathit{Ind} \\ e_1{:}\mathit{Event} \\ e{=}e_1{:}\mathit{Event} \\ \zeta{=}\neg\text{Boolean}{:}\mathit{Scale} \\ c_{\text{BECOME}}{:}\text{become}(c_{\text{RSTATE}},\zeta) \\ c_{\text{RSTATE}}{:}\neg\text{exist}(x,y,\zeta,e_1) \end{bmatrix} \end{bmatrix}$$

- More specified notation of $\lambda x.\text{finish}(x)$
- First line specifies the type of argument: *Ind* (e)
- Second line specifies the $+$MEP
- Third line describes event structure

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Lexicon entry for the head means predicate for Object-oriented RSVC in Yorùbá

$$
\textbf{v1}: \left[\begin{array}{l}
\text{phon:mu} \\
\text{cat:} \left[\begin{array}{lll} \text{head=V} & : & \text{PoS} \\ \text{subj=}\langle \text{sbj-dtr} \rangle & : & \text{list(synsem)} \\ \text{comps=}\langle \text{compl-dtr, vp}_2\text{-dtr} \rangle & : & \text{list(synsem)} \end{array}\right] \\
\text{arg-st=}\langle \text{sbj-dtr, c-dtr, vp}_2\text{-dtr} \rangle\text{:list(synsem)} \\
\text{cont=} \lambda r_1{:}\left[\begin{array}{l} z^s_{11}\text{:IND} \\ e_{12}\text{:Event} \\ \zeta_{12}\text{:Scale} \\ g_{12}\text{:Real} \\ c_{14}\text{:unacc_rel}(z^s_1,\zeta,e_2) \end{array}\right] . \ \lambda r_2{:}[y^c\text{:IND}].\ \lambda r_3{:}[x^s\text{:IND}]. \\
\left[\left[\begin{array}{lll} x^s & : & \text{IND} \\ y^c = r_1.z^s_{11} & : & \text{IND} \\ e{=}e_1 \frown r_1.e_2 & : & \text{Occur} \\ e_1 & : & \text{Event} \\ e_2{=}r_1.e_{12} & : & \text{Event} \\ \zeta{=}r_1.\zeta_{12} & : & \text{Scale} \\ g{=}r_1.g_{12} & : & \text{Real} \\ c_1 & : & \text{drink}_{\text{rsvc}}(r_3.x^s, r_2.y^c, e) \\ c_3 & : & \text{result}(\zeta, g, e_2) \end{array}\right]\right]\text{:TV}
\end{array}\right]
$$

Figure: Lexical entry for a transitive means predicate V_1 selecting an additional result predicate

-Ternary phrasal scheme

- Head-daughter has two arguments on its comps-list: NP_{THEME}, result predicate VP_2
- Mother's comps-list is saturated
- Mother's cont value is the functional application of cont values of the complements to the event record type of V_1
- Result predicate VP_2 is phrasal can contain a direction

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Ternary phrasal scheme

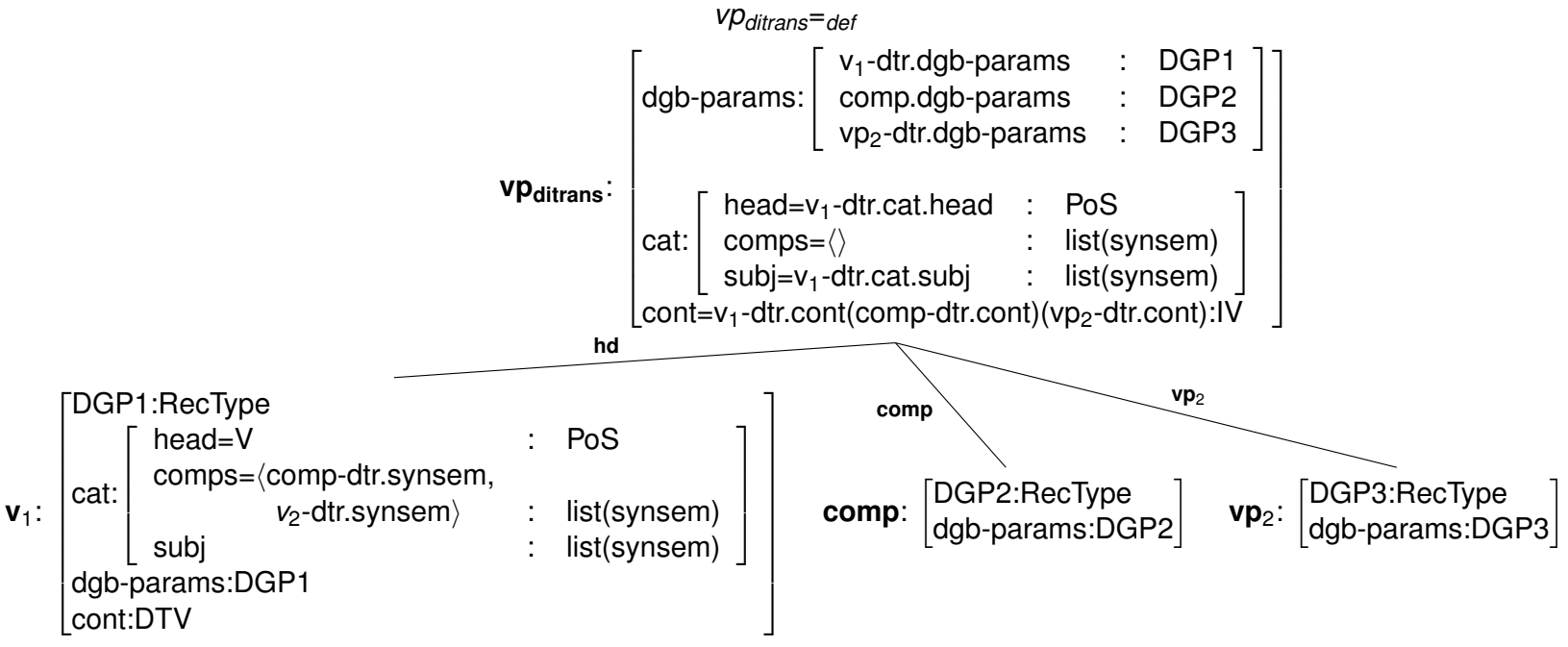
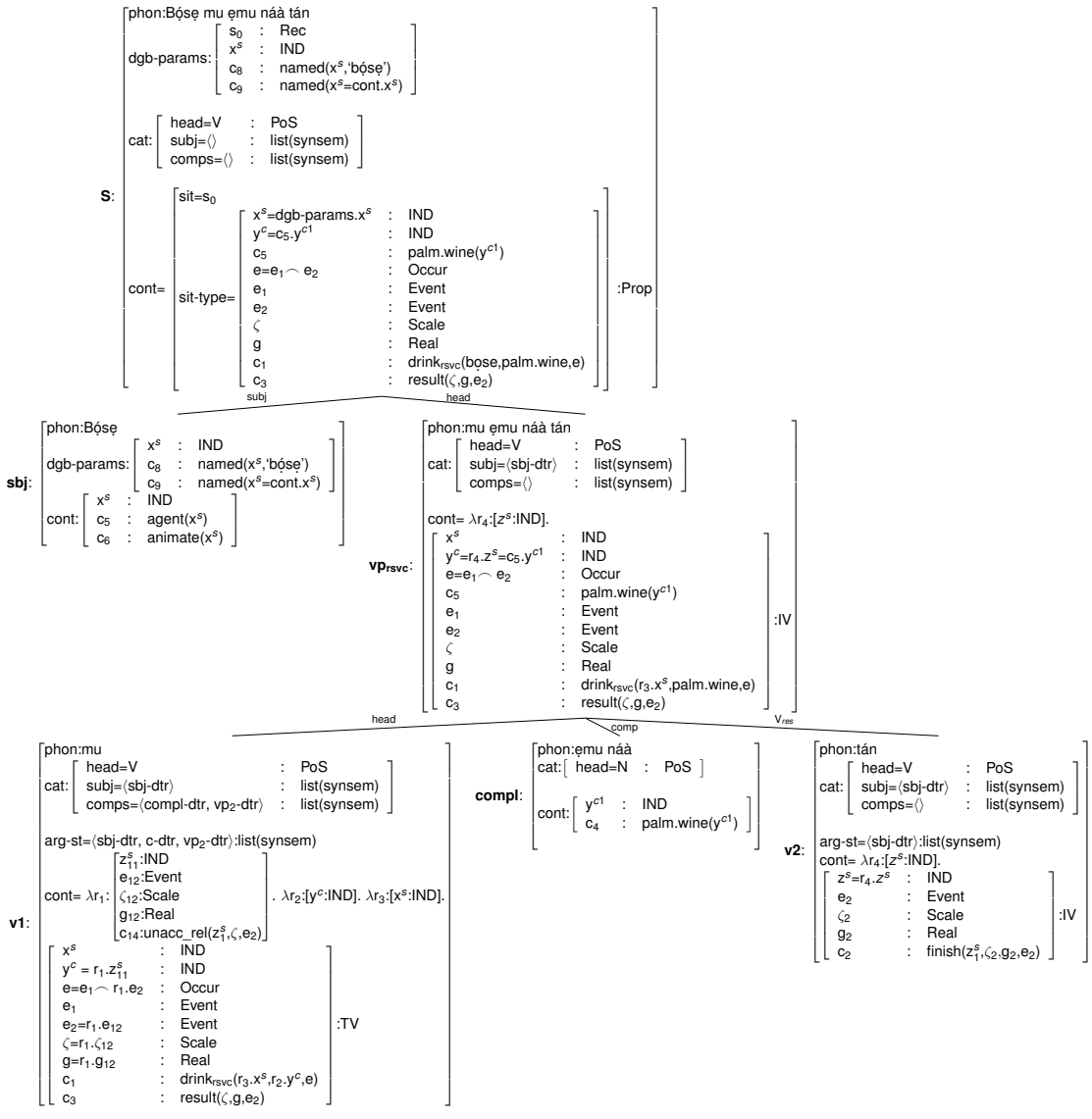
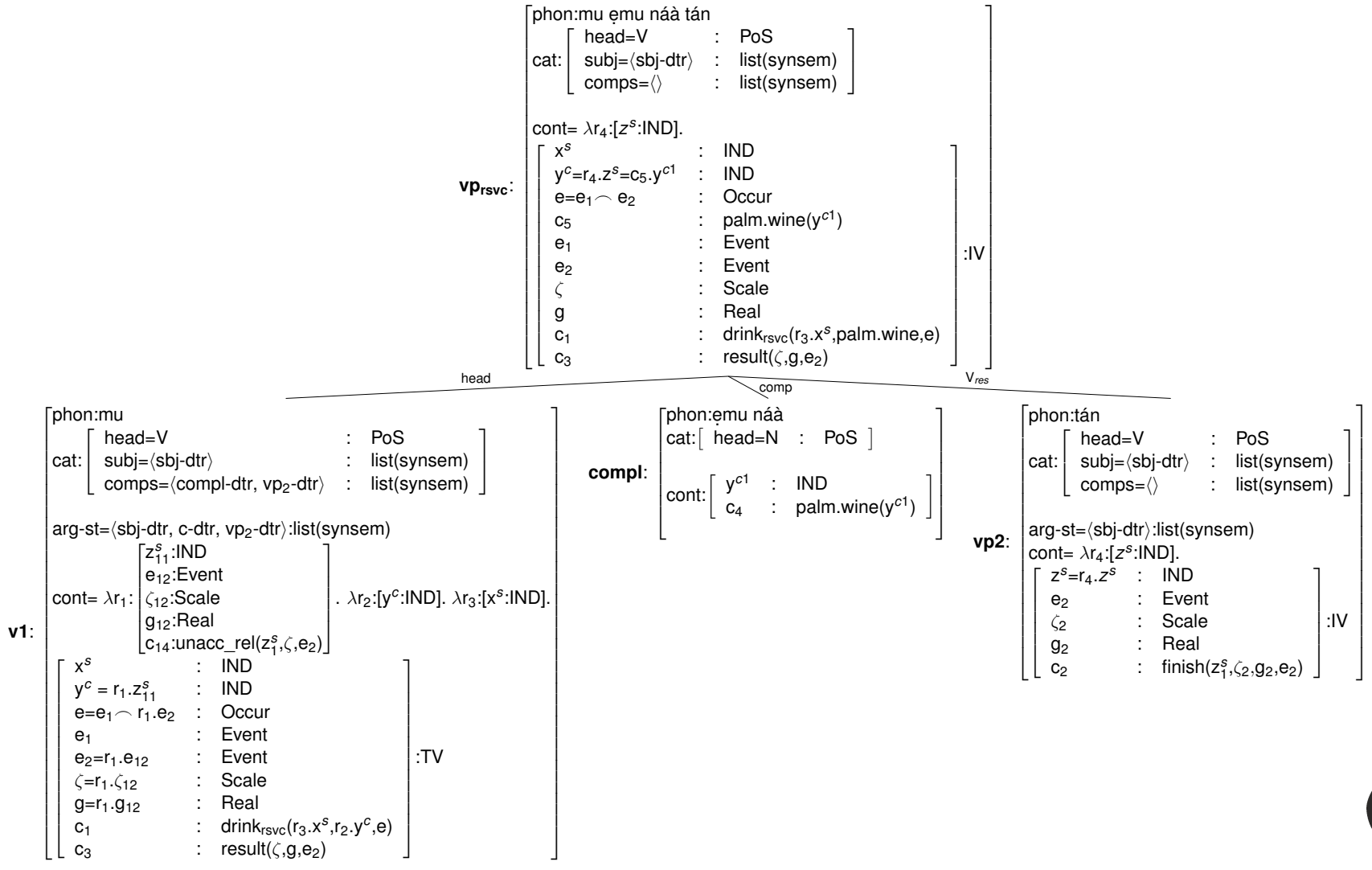


Figure: Phrasal scheme for ditransitive VP

-Object-oriented RSVC in Yorùbá: Full Tree



Object-oriented RSVC in Yorùbá: V_1 + NP_{TH} + VP_2



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-Object-oriented RSVC in Yorùbá: NP_{SUBJ} + VP

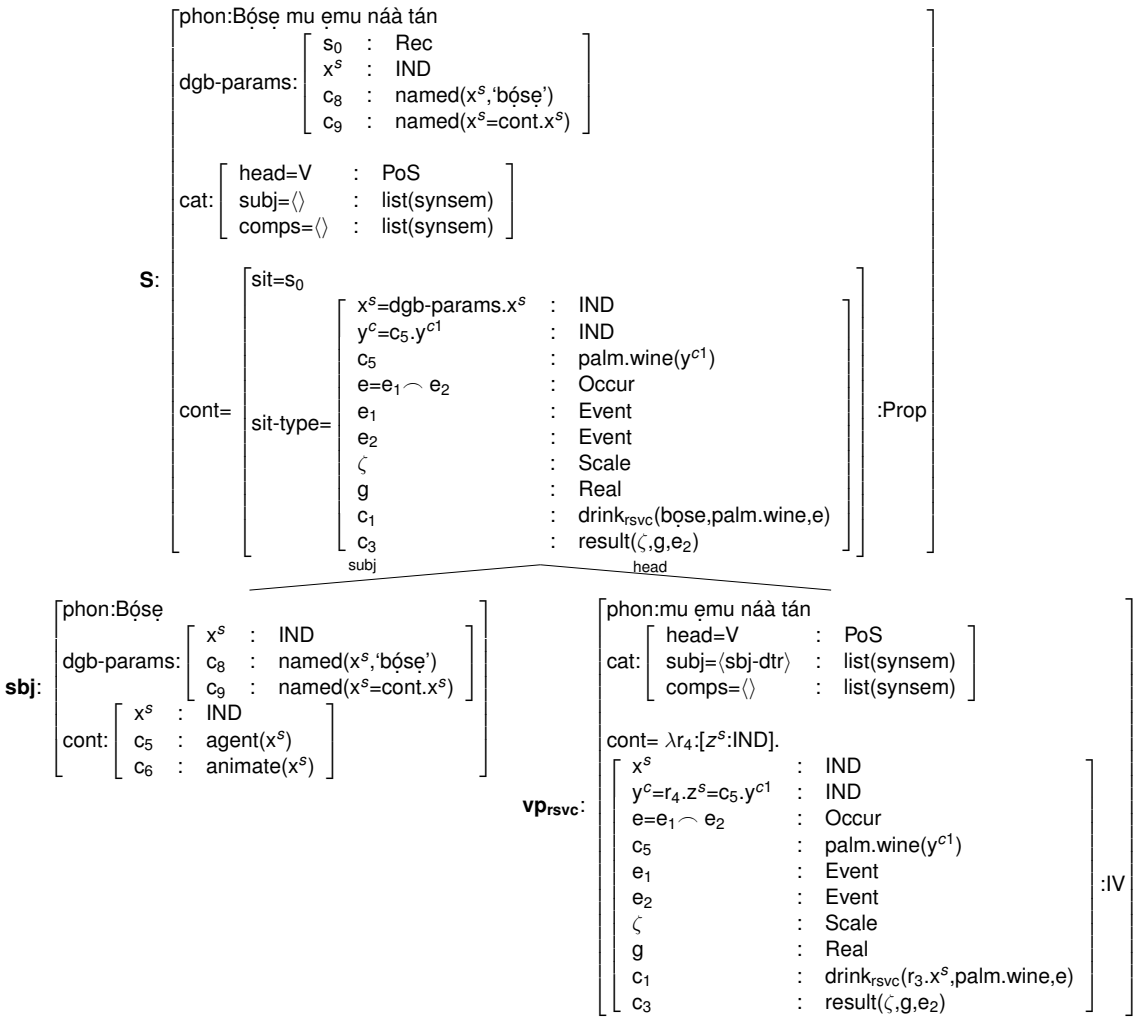


Figure: Object-oriented RSVC in Benue-Kwa: NP_{SUBJ} + VP

Proposition expressed by object-oriented RSVC in Yorùbá

$$
\mathbf{S}:\left[\begin{array}{ll}
\text{phon:Bọ́sẹ̀ mu ẹmu náà tán} & \\
\text{dgb-params:} & \left[\begin{array}{lll} s_0 & : & \text{Rec} \\ x^s & : & \text{IND} \\ c_8 & : & \text{named}(x^s,\text{`bọ́sẹ̀'}) \\ c_9 & : & \text{named}(x^s\text{=cont.}x^s) \end{array}\right] \\
\text{cat:} & \left[\begin{array}{lll} \text{head=V} & : & \text{PoS} \\ \text{subj=}\langle\rangle & : & \text{list(synsem)} \\ \text{comps=}\langle\rangle & : & \text{list(synsem)} \end{array}\right] \\
\text{cont=} & \left[\begin{array}{ll} \text{sit=}s_0 & \\ \text{sit-type=} & \left[\begin{array}{lll} x^s\text{=dgb-params.}x^s & : & \text{IND} \\ y^c\text{=}c_5\text{.}y^{c1} & : & \text{IND} \\ c_5 & : & \text{palm.wine}(y^{c1}) \\ e\text{=}e_1 \frown e_2 & : & \text{Occur} \\ e_1 & : & \text{Event} \\ e_2 & : & \text{Event} \\ \zeta & : & \text{Scale} \\ g & : & \text{Real} \\ c_1 & : & \text{drink}_{\text{rsvc}}(\text{bọse,palm.wine,e}) \\ c_3 & : & \text{result}(\zeta,g,e_2) \end{array}\right] \end{array}\right] : \text{Prop}
\end{array}\right]
$$

Figure: Object-oriented RSVC in Benue-Kwa sentential level

OLD Conclusions

Event structure templates Template Augmentation turns ACTIVITY into ACCOMPLISHMENT takes activity adds LR

- As regards to tense and aspect, Vs in Benue-Kwa have preferred default interpretations (FACTATIVE) but they remain underspecified
  1. (unmarked) eventive predicates prefer a past interpretation
  2. (unmarked) stative predicates prefer a present interpretation
- Following Essegbey (1999, pp. 32, 64–67), Ameka (2008, pp. 135–136, 139–140) and Manfredi, inchoative interpretation is underlying, stative interpretation is derived RSVC can be analysed as a subtype of resultative predicates where the result predicate is realised as VP
- Evidence: VP_2 can be movement predicate, such as Ẹ̀dó *dé* 'fall', Àkán *hwe* 'fall', Efutu *wí* 'fall' or Fɔn *yí* 'go' which preferably are interpreted as events rather than states
- Telicity value is determined by inchoative result predicate


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